

Lecture Notes

Course: Complex Analysis

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Topic: Power Series

Proof 5.3 (Abel-Cauchy-Hadamard's Theorem). Let $\sum a_n(z - z_0)^n$ be a power series and let R be defined as in Hadamard's formula

$$R = \frac{1}{\limsup |a_n|^{1/n}}.$$

Define $w := z - z_0$ and thus

- i) Let $0 < 1/R < \infty$ ($R \in (0, \infty)$), then if $|w| < R$, then by density, there exists a number ρ such that $|w| < \rho < R$ so that $1/\rho > 1/R = \limsup |a_n|^{1/n}$ and thus by the Lemma preceding Cauchy's n th root strong test we have that there exists $K \in \mathbb{N}$ such that $|a_n|^{1/n} < 1/\rho$ for $n \geq K$, this is, $|a_n| < 1/\rho^n$ and hence $|a_n w^n| < (|w|/\rho)^n$ but since $|w|/\rho < 1$ we apply the comparison test to yield that the series $\sum a_n w^n$ is absolutely convergent.

If $R = \infty$, the same reasoning apply to get that the series is absolutely convergent for $|w| < \infty$ (everywhere in $\mathbb{C}$).

- ii) Let $0 < 1/R < \infty$ ($R \in (0, \infty)$), then if $|w| > R$ we apply the same reasoning as before to conclude that $|w| > \rho > R$, $1/\rho < 1/R = \limsup |a_n|^{1/n}$ so there is an infinite set of numbers[†] for which $|a_n|^{1/n} > 1/\rho$ so $|a_n w^n| > (|w|/\rho)^n > 1$ so $|a_n w^n| > 1$ for infinite values of n and thus $\lim(a_n w^n) \neq 0$ so that the series $\sum a_n w^n$ diverges by the n th term test. The proof of statement † goes as follows: Suppose (x_n) is a sequence, then define $u_m = \sup X_m$ so that the sequence (u_m) is decreasing and thus $u_m \geq \lim u_m$ for any $m \in \mathbb{N}$ so $u_m \geq \lim u_m > t$ and $u_m = \sup X_m > t$ so t is not an upper bound of X_m which means that there is some element in the tail $x_n, n > m$ for which

$$x_n > t,$$

but since this n exists for arbitrary m and must satisfy $n > m$, we conclude that there are infinite values of n for which the inequality $x_n > t$ holds.

- iii) Let $0 < r < R$, then there is some ρ such that $r < \rho < R$ and thus $1/\rho > 1/R = \limsup |a_n|^{1/n}$, from the lemma preceding the Cauchy's strong test we have that there is some N for which $|a_n|^{1/n} < 1/\rho$ and hence $|a_n w^n| < (|w|/\rho)^n$ but if $z \in \overline{D}_r(z_0)$, then $|z - z_0| \equiv |w| \leq r$ so $|a_n w^n| < (r/\rho)^n$ and $r/\rho < 1$ so that the series converges uniformly on $\overline{D}_r(z_0)$.

□

Example 5.4. We have $a_n = 1/n!$ so that

$$\lim \frac{a_n}{a_{n+1}} = \lim \frac{(n+1)!}{n!} = \lim(n+1) = \infty,$$

and thus $R = \infty$. We conclude that the series

$$\sum \frac{z^n}{n!}$$

converges absolutely on $\mathbb{C}$ and uniformly on any closed disc.